

HANG ZHANG

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🔗 [zhanghang1989](https://github.com/zhanghang1989) 💼 [zhanghang0704](https://www.linkedin.com/in/zhanghang0704) 📄 [Google Scholar](https://scholar.google.com/citations?user=zhanghang0704)

ABOUT ME

I am Hang Zhang, Head of the VLA Foundation Model Team at XPeng, where I lead the development of [XPeng VLA 2.0](#), the pure-vision, end-to-end Vision-Language-Action foundation model powering XPeng's production driver-assistance system across 300+ cities in China, as well as its overseas vehicles and Robotaxi fleet. I designed [Infini VLA](#), an industry-first streaming VLA architecture that entered production in 2026. I was recognized as one of the [Top 50 Intelligent Driving Developers in China \(2025\)](#).

Previously, I was a Senior Staff Applied Research Scientist at Cruise, where I led the large-scale perception model consolidation effort across detection, segmentation, and 3D perception systems for autonomous driving.

Before Cruise, I was a Research Scientist at Meta, leading the development of a scalable neural architecture optimization platform supporting AI models across Instagram, Portal, and VR devices for applications including person understanding, AR/VR rendering, and ads ranking.

Earlier, I was a Senior Applied Scientist at Amazon AI, working on deep learning and computer vision systems and contributing to the MXNet ecosystem. During that time, we developed ResNeSt, which achieved state-of-the-art results on multiple computer vision benchmarks.

My research has received 12,000+ citations on Google Scholar, and my open-source projects have accumulated 10,000+ GitHub stars.

More about me: [[Homepage](#)] [[GitHub](#)] [[LinkedIn](#)] [[Google Scholar](#)]

EXPERIENCE

XPeng

Dec 2024 - Now

Senior Principal Engineer — Senior Director, Head of VLA Foundation Model

Silicon Valley, CA

- Head of the VLA Foundation Model team (~45 researchers and engineers across Silicon Valley and China) building [XPeng VLA 2.0](#), the pure-vision, end-to-end model behind XPeng's production driver-assistance system in 300+ Chinese cities, overseas vehicles, and Robotaxi; took it from research to production in about a year, delivering three consecutive 2026 releases that roughly halved key safety and intervention metrics across highway, urban, and parking scenarios while raising VLA mileage penetration by over 50%.
- Designed and led [Infini VLA](#), an industry-first streaming VLA architecture shipped in 2026: LLM-style long-sequence training with autoregressive streaming inference over a temporal KV cache, decoupling long-horizon memory from transient reaction on unchanged on-vehicle compute. Wrote the initial implementation and led training, optimization, and integration.
- Advised [TuringViT](#) and the VLM base model, [TuringLLM](#) (20B-A2B MoE for physical AI), and [X-Mind](#) (predictive world model as visual chain-of-thought for driving).

Cruise AI

Sep 2022 - Dec 2024

Senior Staff Applied Research Scientist

San Francisco, CA

- Led efforts in camera major model consolidation on object detection and segmentation in 2023. Led efforts in building a multitask, multiview, multimodality, multiframe, and multiplatform model in Perception.

- Led the development of the Perception model training pipeline, which supports various projects on detection & segmentation, longtailed understanding, lane detection, off-board foundation model, and online distillation.

Meta Reality Lab (Facebook)

Research Scientist

Oct 2020 - Aug 2022

Menlo Park, CA

- Led the development of FBNAS project, a unified pipeline for cross-platform hardware-aware model optimization. FBNAS has been applied to several production models in person understanding on IG, AR/VR applications, and Ads models.
- Developed and open sourced D2Go toolkit, bringing Detectron2 to mobile [[post](#)]
- Research on Open Vocabulary segmentation and efficient architectures, e.g. [OV-Seg](#), [FBNetV5](#), [ScaleViT](#). Co-organized workshop on “Computer Vision for MetaVerse” in ECCV2022.

Amazon AWS AI

Senior Applied Scientist

Jan 2018 - Oct 2020

East Palo Alto, CA

- Led the development of GluonCV toolkit [[link](#)] and AutoGluon toolkit (for AutoML) [[link](#)].
- Led research on large scale vision solution, e.g. [ResNeSt](#), [Bag-of-tricks](#), [CFNet](#), [dynamic SGD](#).
- Organized 3 tutorials on [ICCV19](#), [CVPR20](#) and [ECCV20](#).

Amazon Lab 126 (Internship)

Applied Scientist Intern

May 2017 - Aug 2017

Cupertino, CA

- Developed SoTA semantic segmentation algorithm of [EncNet](#) (Oral paper (~2.1%) in CVPR 2018) [[Talk link on YouTube](#)]

NVIDIA (Internship)

Deep learning Research Intern

May 2016 - Aug 2016

Holmdel, NJ

- Developed an end-to-end deep learning solution for autonomous driving.
- Implemented Torch to Caffe model converter [[GitHub](#)].

EDUCATION

Rutgers University

2013 - 2017

Ph.D. in Electrical and Computer Engineering

Thesis Advisor: Prof. Kristin Dana

Research Interest: Computer Vision

GPA: 3.9/4.0

Southeast University (Nanjing, China)

2009 - 2013

B.S. in School of Automation

Advisor: Prof. Junyang Li

Outstanding Undergraduate Thesis 2013 - School of Automation, Southeast University

PUBLICATIONS

1. Foundation Model Team, XPeng Inc. TuringLLM: Efficiently scaling foundation models toward physical AI. *arXiv preprint arXiv:2608.30567*, 2026
2. Foundation Model Team, XPeng Inc. X-Mind: Efficient visual chain-of-thought via predictive world model for end-to-end driving. *arXiv preprint arXiv:2606.28758*, 2026
3. Foundation Model Team, XPeng Inc. TuringViT: Making SOTA vision transformers accessible to all. *arXiv preprint arXiv:2606.24253*, 2026

4. Runjian Chen, **Hang Zhang**, Avinash Ravichandran, Wenqi Shao, Alex Wong, and Ping Luo. Clap: Unsupervised 3d representation learning for fusion 3d perception via curvature sampling and prototype learning. In *International Conference on Learning Representations (ICLR)*, 2026
5. Feng Liang, Bichen Wu, Xiaoliang Dai, Kunpeng Li, Yinan Zhao, **Hang Zhang**, Peizhao Zhang, Peter Vajda, and Diana Marculescu. Open-vocabulary semantic segmentation with mask-adapted clip. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7061–7070, 2023
6. Bichen Wu, Chaojian Li, **Hang Zhang**, Xiaoliang Dai, Peizhao Zhang, Matthew Yu, Jialiang Wang, Yingyan Lin, and Peter Vajda. Fbnetv5: Neural architecture search for multiple tasks in one run. *arXiv preprint arXiv:2111.10007*, 2021
7. Chaojian Li, Kyungmin Kim, Bichen Wu, Peizhao Zhang, **Zhang, Hang**, Xiaoliang Dai, Peter Vajda, and Yingyan Lin. An investigation on hardware-aware vision transformer scaling. *ACM Transactions on Embedded Computing Systems*, 23(3):1–19, 2024
8. **Zhang, Hang**, Chongruo Wu, Zhongyue Zhang, Yi Zhu, Haibin Lin, Zhi Zhang, Yue Sun, Tong He, Jonas Mueller, R Manmatha, et al. ResNeSt: Split-Attention Networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2736–2746, 2022
9. Yi Zhu, Zhongyue Zhang, Chongruo Wu, Zhi Zhang, Tong He, **Hang Zhang**, R Manmatha, Mu Li, and Alexander Smola. Improving semantic segmentation via efficient self-training. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021
10. **Hang Zhang**, Han Zhang, Chenguang Wang, and Junyuan Xie. Co-occurrent features in semantic segmentation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019
11. Tong He, Zhi Zhang, **Hang Zhang**, Zhongyue Zhang, Junyuan Xie, and Mu Li. Bag of tricks to train convolutional neural networks for image classification. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019
12. Haibin Lin, **Hang Zhang**, Yifei Ma, Zhi Zhang, Sheng Zha, and Mu Li. Elastic distributed training: Learning in the limbo of resources. *arXiv preprint arXiv:1904.12043*, 2019
13. Jian Guo, He He, Tong He, Leonard Lausen, Mu Li, Haibin Lin, Xingjian Shi, Chenguang Wang, Junyuan Xie, Sheng Zha, Aston Zhang, **Hang Zhang**, Zhi Zhang, Zhongyue Zhang, and Shuai Zheng. Gluoncv and gluonnlp: Deep learning in computer vision and natural language processing. *arXiv preprint arXiv:1907.04433*, 2019
14. Parneet Kaur, **Hang Zhang**, and Kristin Dana. Photo-realistic facial texture transfer. In *Winter Conference on Applications of Computer Vision (WACV)*, 2019
15. **Hang Zhang**, Kristin Dana, Jianping Shi, Zhongyue Zhang, Xiaogang Wang, Amrith Tyagi, and Amit Agrawal. Context encoding for semantic segmentation. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018 (**oral**)
16. Jia Xue, **Hang Zhang**, and Kristin Dana. Deep texture manifold for ground terrain recognition. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018
17. **Hang Zhang** and Kristin Dana. Multi-style generative network for real-time transfer. *European Conference of Computer Vision Workshops(ECCVW)*, 2018
18. **Hang Zhang**. *Reflectance and texture encoding for material recognition and synthesis*. PhD thesis, Rutgers University-School of Graduate Studies, 2017
19. **Hang Zhang**, Jia Xue, and Kristin Dana. Deep ten: Texture encoding network. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, July 2017

20. Jia Xue, **Hang Zhang**, Kristin Dana, and Ko Nishino. Differential angular imaging for material recognition. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, July 2017
21. **Hang Zhang**, Kristin Dana, and Ko Nishino. Friction from reflectance: Deep reflectance codes for predicting physical surface properties from one-shot in-field reflectance. In *European Conference on Computer Vision (ECCV)*, pages 808–824. Springer, 2016
22. **Hang Zhang**, Kristin Dana, and Ko Nishino. Reflectance hashing for material recognition. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3071–3080, 2015

ACADEMIA SERVICES

Workshop and Tutorial Organizer

<i>European Conference on Computer Vision (ECCV)</i>	Tel Aviv, 2022
Computer Vision for MetaVerse	
<i>European Conference on Computer Vision (ECCV)</i>	Glasgow, 2020
From HPO to NAS: Automatic Deep Learning.	
<i>IEEE Conference on Computer Vision and Pattern Recognition (CVPR)</i>	Seattle, 2020
From HPO to NAS: Hands-on Tutorial on Automatic Deep Learning.	
<i>IEEE International Conference on Computer Vision (ICCV)</i>	Seoul, 2019
Everything You Need to Know to Reproduce SOTA Deep Learning Models:	
Hands-on Tutorial for Training SOTA Computer Vision Models.	
<i>Amazon Machine Learning Conference (AMLC)</i>	Seattle, 2018
CNNs for Semantic Segmentation.	

Reviewer for Journals:

IEEE TPAMI, IEEE TBioCAS, Computer Vision and Image Understanding (CVIU)

Program Committee and Reviewer for Conferences:

CVPR (2018–2021), ICCV (2019–2021), ECCV (2018–2020), NeurIPS (2020–2021), WACV (2018–2019), SIGGRAPH (2018)